WEEK 3 · LECTURE L1 · UNIT I

Data Literacy and Statistical Thinking for HR

How to read an HR number without being fooled: populations and samples, rates and counts, base rates, and correlation versus causation.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Distinguish population, sample and self selection, and explain why selection breaks HR data.
2. Choose correctly between counts and rates, percent and percentage points, mean and median.
3. Explain base rate neglect, Simpson’s paradox, survivorship, regression to the mean and the ecological fallacy with HR examples.
4. Evaluate a causal claim using the counterfactual view and the Bradford Hill considerations.
5. Use levels of measurement, spread and effect size correctly, and place yourself on the data literacy levels.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 1 and CLO 3.

COMMIT BEFORE THEORY

Commit to an answer first

Plant A had 60 resignations last year and Plant B had 30. Which plant has the bigger retention problem?

A

Plant A, because it had twice as many resignations

B

Cannot tell without headcount: a rate, not a count, compares plants

C

Plant B, because smaller numbers hide problems

D

Neither, because resignations are normal

EXAM LENS

The answer is B. If Plant A has 1,200 staff and Plant B 300, the rates are 5% and 10%: twice the count, half the rate.

DATA LITERACY, ONE OF THREE

Who is actually in this number?

Population

Everyone you mean to describe. HR files rarely contain it: leavers drop out, contractors are excluded, new joiners are not yet loaded.

Sample

The subset you actually have. It supports a claim about the population only if you know how it was selected. A stratified sample carries shares across; counts must be scaled up, and the scaling stated.

Self selection

The dangerous case. Survey respondents chose to answer; app users chose to log in. Those missing are missing for a reason linked to what you measure. A low response rate describes the respondents, not the workforce.

EXAM LENS

Ask who is missing before you ask what the number says. Selection, not sampling error, is what usually breaks HR data. A sample share also carries sampling error: a 95% interval says how far the true figure could reasonably be.

DATA LITERACY, TWO OF THREE

A count tells you size; a rate tells you risk

Claim	What it is	Where it misleads
“Store operations lost the most people”	A count	The largest department almost always loses the most people
“Store operations has the highest attrition rate”	A rate: leavers over a base	Only as good as the base: which headcount, when, including whom?
“Attrition rose from 31% to 38%”	A rate over time	A falling denominator raises a rate without anyone extra leaving
“Team X has 33% attrition”	A small base rate	4 exits in a team of 12 is 33%; one more is 42%. Report the count beside the rate

EXAM LENS

Always state three things: the definition (which exits), the period, and the denominator.

DATA LITERACY, THREE OF THREE

Points, scale, and an average that hides the story

Points, not percent

A rate rising from 20% to 25% rises by 5 percentage points, which is a 25 percent increase: the second sounds five times larger. Say which you mean.

Sample or full scale

If a file holds a one in eight sample, a Board figure must be scaled up, and the scaling stated. Divide by the people actually in the table.

Means hide spreads

An average digital literacy of 3 out of 5 may describe almost nobody if metro staff cluster high and small town staff low. Segment by location, grade, device and contract before concluding; report the share low and high.

EXAM LENS

Read every number aloud with its unit and its base: “a 5 percentage point rise, among 300 workers”.

Levels of measurement, center, spread and significance

Level	Examples	What you may compute
Nominal	Department, gender	Counts and proportions only
Ordinal	Ratings, grade bands	Medians and distributions, not means
Interval	Equal steps, arbitrary zero	Differences, not ratios
Ratio	Salary, tenure	All operations

FROM THE TEACHING NOTE

Averaging a 1 to 5 rating treats unequal steps as equal: report the median or distribution. Pay is ratio and right skewed, so report the median beside the mean. Two groups with the same mean can differ sharply in spread: use the standard deviation, and the coefficient of variation across scales. With large samples trivial effects become “significant”: report the effect size.

THEORY IN DEPTH · TVERSKY AND KAHNEMAN (1974)

We neglect base rates

Of 1,000 staff	Flagged	Not flagged	Total
Leave (5%)	40	10	50
Stay (95%)	190	760	950
Total	230	770	1,000

WORKED EXAMPLE · COURSE FRAMING

A flag that is “80% accurate” catches 40 of 50 leavers but also 190 of 950 stayers: only about 17% of flagged staff actually leave (40 ÷ 230). When the event is rare, most flags are false alarms.

The core idea

People judge probability by how well a case fits a story (representativeness), neglecting how common the outcome is. The fix: start from the base rate, then adjust.

Natural frequencies

Base rates are used far better when framed as counts out of 100 or 1,000 (Gigerenzer and Hoffrage, 1995). Show the arithmetic this way in answers.

THEORY IN DEPTH · BICKEL, HAMMEL AND O’CONNELL (1975)

Simpson’s paradox: the aggregate can mislead

The core idea

Berkeley graduate admissions appeared to favor men overall (about 44% of men admitted against 35% of women).

How it works

Department by department, women were admitted at similar or slightly higher rates; women had applied more to competitive departments. A pattern can appear, vanish or reverse when groups are combined.

Limits and critique

The right level of analysis depends on the causal question, not on the data alone.

IN PRACTICE

An organization wide gender gap in promotions or offers must be checked within each role, grade and location before concluding.

EXAM LENS

Use this case whenever a question asks you to evaluate an aggregate HR statistic.

BEFORE CLAIMING CAUSATION

Four things a correlation can be

Explanation	Meaning	HR example
A causes B	The claim you want; needs a mechanism, time order and ideally variation you did not choose	Training raises performance
B causes A	Reverse causation	Engaged staff volunteer for training
C causes both	A confounder	Grade drives device access, literacy and app usage at once
Chance, or selection	The pattern lives in noise, or in how the file was built	Only active users appear in usage logs

EXAM LENS

A strong answer names and rules out the rival explanations before accepting a causal claim. A model that leans on a variable such as age has found a proxy, not a driver.

THEORY IN DEPTH · RUBIN (1974); HOLLAND (1986)

Causation means comparing with the counterfactual

The core idea

The effect of training on a person is the difference between their outcome with and without it.

How it works

We see only one outcome (Holland’s “fundamental problem of causal inference”), so we compare groups; random assignment makes the groups comparable.

Limits and critique

Without randomization, hidden differences between groups can create a false effect.

IN PRACTICE

Employees who chose a course may differ from those who did not; compare like with like before crediting the course.

EXAM LENS

Holland’s phrase is precise and worth using in critical answers.

THEORY IN DEPTH · BRADFORD HILL (1965)

Nine considerations for judging a causal claim

The core idea

When experiments are impossible, weigh several kinds of evidence for causation.

How it works

Strength, consistency, specificity, temporality, biological gradient (dose response), plausibility, coherence, experiment and analogy.

Limits and critique

Only temporality is strictly necessary; the rest are guides, not a checklist to add up.

IN PRACTICE

Does higher engagement come before lower attrition, and do more engaged teams leave less? Test temporality first.

EXAM LENS

Hill called these “viewpoints”, not criteria. Applying two or three carefully beats listing all nine.

COMMON HR DATA FALLACIES

Six traps to catch

Survivorship

Studying only those who stayed hides why others left; usage logs sample active users; exit interviews hear only leavers.

Regression to the mean

The worst units this quarter usually improve next quarter anyway. Credit only change beyond that drift.

Ecological fallacy

Inferring individuals from group averages: a highly engaged store can hide a leaver.

Small numbers

A rate on 12 people swings wildly; report the count beside the rate.

Moving definitions

A metric that changes definition midseries, or a cherry picked start date, creates a false trend.

Correlation as cause

Training and performance move together; which caused which?

COURSE FRAMING

Most HR analytics errors are reasoning errors, not calculation errors.

WORKED

Before believing “engagement rose 5%”

1 · Percent or points?

A 5 point rise and a 5 percent rise are different claims.

2 · From what base?

Who responded, and is the response rate comparable?

3 · Same instrument?

Same questions and scale as last time?

4 · Who responded?

If frontline staff are missed, a rise may only reflect who answered. Did it rise in every segment, and beyond sampling error?

COURSE FRAMING

The honest reformulation: “Among the respondents, mostly corporate staff, the average score rose by X points; we cannot say whether frontline engagement changed.”

DATA LITERACY

Four competencies and five levels

Four competencies

Reading data, working with data, analyzing data and communicating with data.

Five levels

1 Awareness · 2 Comprehension (charts, averages, quality issues) · 3 Application (pivot tables, HR metrics, correlation versus causation) · 4 Analysis (regression, significance, study design) · 5 Innovation (experiments, leading data driven change).

Targets by role

HR professionals at 3 to 4; line managers 2 to 3; executives 3; analysts 4 to 5. The teaching note reports most HR professionals at levels 1 and 2.

Course’s Unit 2 teaching note, drawing on the Qlik Data Literacy Project (2018).

APPLY

Three claims from an HR report

Teaching case (Course framing)

A Hyderabad IT firm’s HR report says: (1) the Pune office has the worst attrition, with 90 exits against Hyderabad’s 70; (2) a new flight risk model is “85% accurate”, though 6% of staff leave each year; (3) women are promoted at a lower rate than men overall, so managers are biased.

Work it through

1. What do you need before accepting claim 1?
2. Why might claim 2 be misleading? Use base rates.
3. What should be checked before accepting claim 3?
4. Name one more fallacy to watch for in such a report.

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
A bigger count means a bigger problem	Compare rates with stated denominators
An accurate model gives mostly correct flags	With a low base rate, most flags can be false alarms
An overall gap proves bias at every level	Simpson’s paradox: check the groups before concluding
A 5% rise and a 5 point rise are the same	Percent change and percentage points differ
The average describes the typical employee	Means hide spreads; ordinal data need medians

COMPARE AND EVALUATE

Six lenses for reading an HR number

Lens	Question it answers	Watch out for
Population, sample, selection	Who is in this number?	Selection hides the people you most need
Rates and denominators	How big is the risk, not the group?	Small bases swing
Base rates (Tversky and Kahneman)	How surprising is this case?	Rare events breed false alarms
Simpson’s paradox (Bickel et al.)	Does the total hold in the groups?	The right level depends on the question
Counterfactual (Rubin; Holland)	What would have happened otherwise?	Needs comparable groups
Bradford Hill	Is the causal case strong?	Viewpoints, not a checklist

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

An Indian bank’s HR head concludes from aggregate data that women are under promoted and that a new flight risk tool is reliable. Using rates, base rates and Simpson’s paradox, evaluate both conclusions and state what further analysis is needed. **20 marks**

A strong answer: distinguishes counts from rates with the right denominators; applies base rates to the flight risk claim with a worked calculation; explains Simpson’s paradox and the breakdowns to check; states what the data cannot show, such as cause.

“HR data rarely lie, but they often mislead.” Discuss with reference to selection, base rates and Simpson’s paradox. **20 marks**

A strong answer: explains each trap with an HR example; shows the base rate arithmetic; uses the Berkeley case; concludes with practical safeguards.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

A firm finds that employees who complete training have lower attrition and concludes training reduces attrition. Critically evaluate this conclusion. **20 marks**

A strong answer: lays out reverse causation, confounding and selection; applies the counterfactual view and Bradford Hill; proposes a design that could test the claim.

Distinguish a percentage change from a change in percentage points, with an example. **10 marks**

A strong answer: gives a correct worked example and explains why the distinction matters in HR reporting.

Why should a 1 to 5 appraisal rating be summarized by its median rather than its mean? **10 marks**

A strong answer: explains ordinal measurement and unequal steps; offers the distribution as the better report.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Every rate has a numerator and a denominator.**
   State the definition, the period and the base before comparing.

2. **Ask who is missing.**
   Selection, survivorship and response rates decide what a number can say.

3. **Means hide spreads.**
   Split by segment, and use medians for ordinal or skewed data.

4. **Correlation is a clue, not a cause.**
   Rule out reverse causation, confounding, chance and selection.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Kavanagh, M. J., & Johnson, R. D. (2018). Chapter 2 (data, information and knowledge; warehouses, business intelligence and data mining) and Chapter 14 (HR metrics and workforce analytics).
- Angrave, D., et al. (2016). HR and analytics: Why HR is set to fail the big data challenge. *Human Resource Management Journal, 26*(1), 1 to 11.

NOTE

The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Angrave, D., Charlwood, A., Kirkpatrick, I., Lawrence, M., & Stuart, M. (2016). HR and analytics: Why HR is set to fail the big data challenge. *Human Resource Management Journal, 26*(1), 1 to 11. https://doi.org/10.1111/1748-8583.12090
- Bickel, P. J., Hammel, E. A., & O'Connell, J. W. (1975). Sex bias in graduate admissions: Data from Berkeley. *Science, 187*(4175), 398 to 404. https://doi.org/10.1126/science.187.4175.398
- Gigerenzer, G., & Hoffrage, U. (1995). How to improve Bayesian reasoning without instruction: Frequency formats. *Psychological Review, 102*(4), 684 to 704. https://doi.org/10.1037/0033-295X.102.4.684
- Hill, A. B. (1965). The environment and disease: Association or causation? *Proceedings of the Royal Society of Medicine, 58*(5), 295 to 300. https://doi.org/10.1177/003591576505800503
- Holland, P. W. (1986). Statistics and causal inference. *Journal of the American Statistical Association, 81*(396), 945 to 960. https://doi.org/10.1080/01621459.1986.10478354
- Kavanagh, M. J., & Johnson, R. D. (Eds.). (2018). *Human resource information systems* (4th ed.). Sage.
- Qlik Data Literacy Project. (2018). Data literacy levels, as cited in the course's Unit 2 teaching note.
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology, 66*(5), 688 to 701. https://doi.org/10.1037/h0037350
- Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science, 185*(4157), 1124 to 1131. https://doi.org/10.1126/science.185.4157.1124